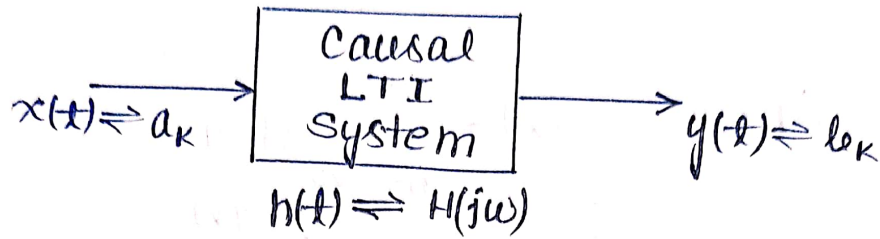


SAS QUIZ-5 SOLUTION

Sol(1):



$$2 \frac{d}{dt} y(t) + 3 y(t) = x(t) \quad (\text{Given}) \quad (1)$$

$$\text{Let } x(t) = e^{j\omega t} \quad (2)$$

$$\text{then } y(t) = H(j\omega) \cdot e^{j\omega t} \quad (3)$$

$$\text{Now, } 2 \frac{d}{dt} \{ H(j\omega) \cdot e^{j\omega t} \} + 3 \{ H(j\omega) \cdot e^{j\omega t} \} = e^{j\omega t}$$

$$2 H(j\omega) \cdot (j\omega) \cdot e^{j\omega t} + 3 H(j\omega) e^{j\omega t} = e^{j\omega t}$$

$$2 j\omega H(j\omega) + 3 H(j\omega) = 1$$

$$H(j\omega) = \left(\frac{1}{3 + j2\omega} \right) = \text{Frequency response of the system}$$

If $x(t)$ is periodic with 'T' then — (4) $\rightarrow$ (2 Marks)

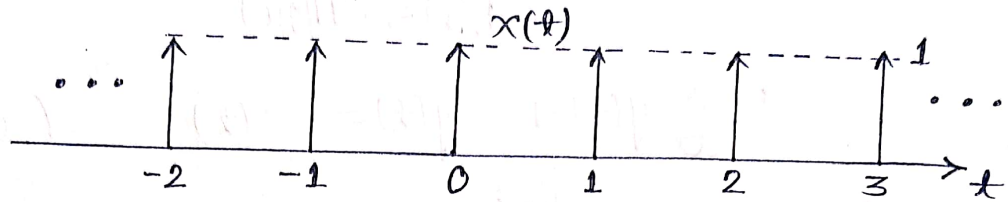
$$x(t) = \sum_{k=-\infty}^{\infty} a_k \cdot e^{jk\omega_0 t} \quad (5)$$

$$\begin{aligned} \text{Therefore, } y(t) &= x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau \\ &= \int_{-\infty}^{\infty} h(\tau) \cdot \sum_{k=-\infty}^{\infty} a_k \cdot e^{jk\omega_0(t-\tau)} d\tau \\ &= \sum_{k=-\infty}^{\infty} a_k \cdot e^{jk\omega_0 t} \cdot \int_{-\infty}^{\infty} h(\tau) \cdot e^{-jk\omega_0 \tau} d\tau \\ &= \sum_{k=-\infty}^{\infty} a_k \cdot e^{jk\omega_0 t} \cdot H(jk\omega_0) \\ &= \sum_{k=-\infty}^{\infty} a_k \cdot H(jk\omega_0) \cdot e^{jk\omega_0 t} \quad (6) \end{aligned}$$

By eqⁿ(6), Fourier series coefficient of $y(t)$ -

$$b_k = a_k \cdot H(jk\omega_0) \quad \text{--- (7)} \quad \rightarrow (2 \text{ Marks})$$

Sol 1(a)÷ Given - $x(t) = \sum_{n=-\infty}^{\infty} \delta(t-n) \Rightarrow a_k$



$$\begin{aligned} \therefore a_k &= \frac{1}{T} \int_T x(t) \cdot e^{-jk\omega_0 t} \cdot dt \\ &= \int_{-0.5}^{0.5} \delta(t) e^{-jk(2\pi)t} \cdot dt \end{aligned}$$

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T_0} \\ &= \frac{2\pi}{1} = 2\pi \end{aligned}$$

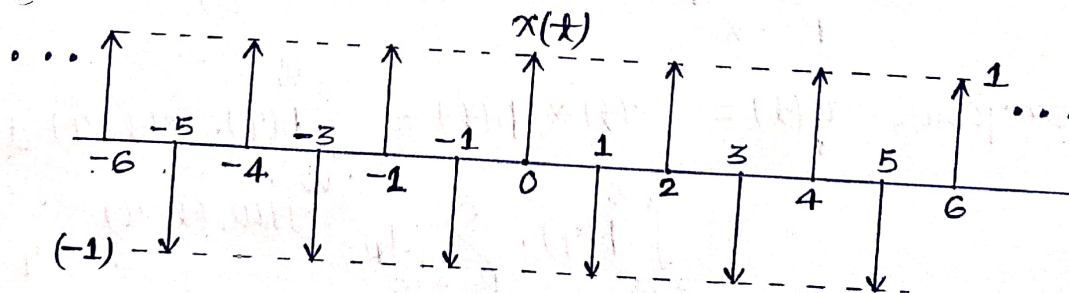
$$= 1$$

$\rightarrow (1.5 \text{ Marks})$

$$\therefore b_k = a_k \cdot H(jk\omega_0)$$

$$= 1 \cdot \frac{1}{3 + j2(k\omega_0)} = \frac{1}{3 + j2k\omega_0} = \left(\frac{1}{3 + j4k\pi} \right)$$

Sol 1(b)÷ Given - $x(t) = \sum_{n=-\infty}^{\infty} (-1)^n \cdot \delta(t-n) \Rightarrow a_k$ $\rightarrow (1.5 \text{ Marks})$



$$\begin{aligned} \therefore a_k &= \frac{1}{T} \int_T x(t) \cdot e^{-jk\omega_0 t} \cdot dt \\ &= \frac{1}{2} \int_{-0.5}^{1.5} x(t) \cdot e^{-jk\pi t} \cdot dt \end{aligned}$$

$$\begin{aligned} T_0 &= 2 \\ \therefore \omega_0 &= \frac{2\pi}{T_0} = \pi \end{aligned}$$

$$= \frac{1}{2} \int_{-0.5}^{0.5} \delta(t) e^{-jk\pi t} \cdot dt - \frac{1}{2} \int_{0.5}^{1.5} \delta(t-1) e^{-jk\pi t} \cdot dt$$

$$a_k = \left(\frac{1}{2} - \frac{1}{2} e^{-jk\pi} \right)$$

→ (1.5 Marks)

$$\therefore b_k = a_k \cdot H(jk\omega_0)$$

$$= \left\{ \frac{1}{2} - \frac{1}{2} e^{-jk\pi} \right\} \cdot \frac{1}{3 + j2(k\omega_0)}$$

$$= \frac{1}{2} (1 - e^{-jk\pi}) \cdot \left(\frac{1}{3 + j2k\pi} \right)$$

$$= \frac{1}{2} (1 - (-1)^k) \left(\frac{1}{3 + j2k\pi} \right)$$

→ (1.5 Marks)